THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET A

Chapters 5 & 6: **Circles, Measuring Space: Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a circle of radius 10 cm, a chord is 6 cm from the centre. The length of the chord is: 1

(a) 18 cm **(b)** 16 cm **(c)** 8 cm **(d)** 14 cm

2. In a circle of radius 13 cm, a chord is 12 cm from the centre. The length of the chord is: 1

(a) 8 cm **(b)** 10 cm **(c)** 12 cm **(d)** 5 cm

3. A chord of length 30 cm is at a distance of 8 cm from the centre of a circle. The radius of the circle is: 1

(a) 23 cm **(b)** 17 cm **(c)** 18 cm **(d)** 30 cm

4. The diameter of a circle is 26 cm and a chord is 24 cm long. The distance of the chord from the centre is: 1

(a) 25 cm **(b)** 1 cm **(c)** 13 cm **(d)** 5 cm

5. An arc subtends an angle of 70° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 70° **(b)** 145° **(c)** 35° **(d)** 140°

6. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 21 cm is: 1

(a) 66 cm **(b)** 132 cm **(c)** 264 cm **(d)** 1386 cm

7. The perimeter of a circle is 88 cm. Taking $\pi = \frac{22}{7}$, its radius is: 1

(a) 7 cm **(b)** 28 cm **(c)** 14 cm **(d)** 15 cm

8. The length of an arc subtending 90° at the centre of a circle of radius 21 cm ($\pi = \frac{22}{7}$) is: 1

(a) 16.5 cm **(b)** 132 cm **(c)** 33 cm **(d)** 66 cm

9. The length of an arc subtending 30° at the centre of a circle of radius 42 cm ($\pi = \frac{22}{7}$) is: 1

(a) 11 cm **(b)** 22 cm **(c)** 44 cm **(d)** 264 cm

10. By Heron's formula, the area of a triangle with sides 8 cm, 15 cm and 17 cm is: 1

(a) 120 cm^2 **(b)** 66 cm^2 **(c)** 54 cm^2 **(d)** 60 cm^2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In a circle of radius 12 cm, the chord AB subtends 60° at the centre O. Find AB. 2

12. A circle has radius 13 cm. Two chords of length 10 cm each are drawn. Find their distance from the centre and state the relation between the distances. 2

13. Use Heron's formula to find the area of a triangle of sides 3 cm, 4 cm and 5 cm and verify with $\frac{1}{2}bh$. 2

14. The area of a parallelogram is 96 cm^2 and its base is 12 cm. Find its height. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In cyclic quadrilateral ABCD, $\angle A = (4x - 4)^\circ$ and $\angle C = (2x + 16)^\circ$. Find x and both angles. 3

16. The circumferences of two circles are in the ratio 5 : 4. Find the ratio of their areas. 3

17. Show that a cyclic parallelogram is a rectangle. 3

OR

ABCD is a cyclic trapezium with $AB \parallel CD$. Show that its non-parallel sides AD and BC are equal. 3

18. A circular park has perimeter 88 m. Find its area and the cost of fencing it at ₹12 per metre ($\pi = \frac{22}{7}$). 3

OR

A circular ring has outer radius 14 cm and inner radius 7 cm. Find its area ($\pi = \frac{22}{7}$). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A round table has centre O and radius 60 cm. Three diners sit at A, B, C on its edge with $\angle BAC = 40^\circ$. 5

(i) Find $\angle BOC$.

(ii) Find $\angle OBC$.

(iii) A fourth diner sits at D on the same arc as A; find $\angle BDC$.

(iv) If M is the midpoint of BC and $OM = 36$ cm, find BC.

OR

Prove that if two chords of a circle are equidistant from the centre, they are equal; then find the length of each chord if the distance is 12 cm and the radius is 13 cm. 5

20. Prove, using the inscribed and circumscribed regular hexagons of a circle of radius r, that $3 < \pi < 2\sqrt{3}$. Why is Archimedes' extension to 96 sides useful? 5

OR

Two congruent circles of radius r with centres A and B each pass through the other's centre and meet at C and D. 5

(i) Prove that $\triangle ABC$ is equilateral.

(ii) Find $\angle CAD$.

(iii) Find the perimeter of the outer boundary of the figure.

END OF QUESTION PAPER — **Total: 40 Marks**